# Correlation

## Starter



1) The table shows the scores out of $10$ on a maths test and on a physics test for $7$ students. Show this information on a scatter diagram.

| **Maths** | 6 | 7 | 7 | 8 | 9 | 9 | 10 |
|---|---|---|---|---|---|---|---|
| **Physics** | 9 | 7 | 6 | 7 | 5 | 4 | 5 |

1

2) A straight line has equation $y = 0.34 - 3.21x$.

Write down

a) The gradient of the line

2a

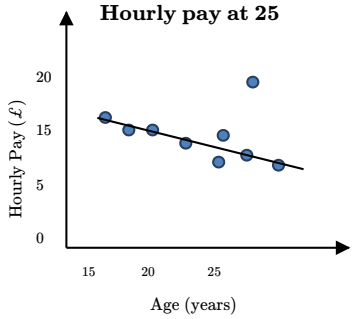
b) The $y$-intercept of the line

2b

Previously we have only considered one variable at a time. When we introduce a second variable (e.g. height with age), **we might want to consider the relationship between them**.

This is a short chapter!

"Describe the type of correlation."




"The daily mean windspeed, $w$ knots, and the daily maximum gust, $g$ knots, were recorded. The equation of the regression line of $g$ on $w$ for these 15 days is $g = 7.23 + 1.82w$.

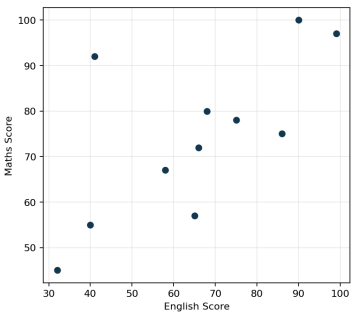
(a) Given an interpretation of the value of the gradient of this regression line.

(b) Justify the use of a linear regression line in this instance."

**Changes since the old 'S1' syllabus:**
This chapter has been scaled back significantly since the S1 'Correlation' and 'Regression' chapters. You no longer need to determine the equation of the line of best fit (the regression line), or calculate measures of correlation, but merely have to interpret an equation already given or the limitations of estimates made or comment on the suitability of a linear regression model.



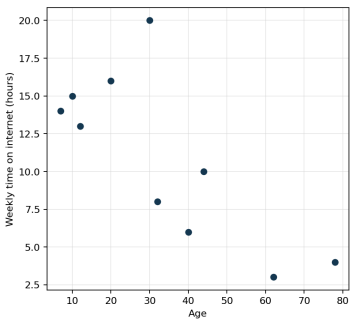
Correlation gives the **strength of the relationship** (and the type of relationship) between two variables. Data with two variables is known as **bivariate data**.



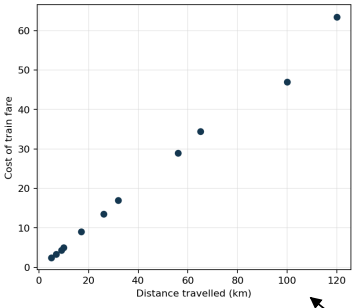

Type of correlation:

[ ] [ ] correlation

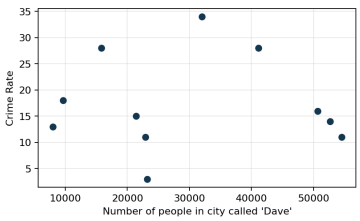
strength type




[ ] [ ] correlation




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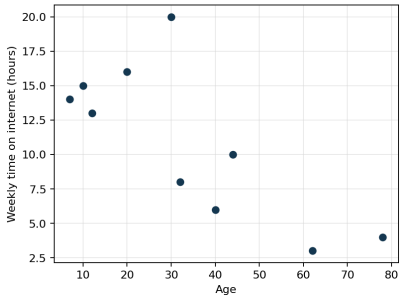



[ ] correlation

The vertical-axis variable usually **depends** on the horizontal-axis value. For this reason distance would be the **independent/explanatory variable** and cost the **dependent/response variable**.

## Important Point 1

To **interpret** the correlation between two variables is to give a worded description in the context of the problem.



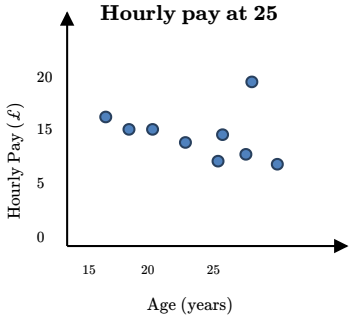

a) State the correlation shown.

b) Describe/interpret the relationship between age and weekly time on the internet.

## Important Point 2

[Textbook] Two variables have a **causal relationship** if a change in one variable directly causes a change in the other. Just because two variables show correlation it does not necessarily mean that they have a causal relationship.

Hideko was interested to see if there was a relationship between what people earn and the age which they left education or training. She says her data supports the conclusion that more education causes people to earn a lower hourly rate of pay. Give one reason why Hideko's conclusion might not be valid.






3. Eight students estimated the mass of a bag of sweets, in grams. First they estimated without touching the bag. They then picked it up and estimated the mass again.

| Student | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| Without touching (g) | 25 | 18 | 32 | 27 | 21 | 35 | 28 | 30 |
| Holding the bag (g) | 16 | 11 | 20 | 17 | 15 | 26 | 22 | 20 |

(a) Draw a scatter diagram to represent the data.

a

(b) Describe and interpret the correlation between the two variables.

b

4. Donal investigated whether house values were related to internet speed, measured by the time taken to download a 100 megabyte file. The table shows his results.

| Time taken (s) | 5.2 | 5.5 | 5.8 | 6.0 | 6.8 | 8.3 | 9.3 | 13 | 13.6 | 16.0 |
|---|---|---|---|---|---|---|---|---|---|---|
| House value (£1000s) | 300 | 310 | 270 | 200 | 230 | 205 | 208 | 235 | 175 | 180 |

(a) Draw a scatter diagram to represent the data.

a

(b) Describe the type of correlation.

b

(c) Donal says a slow connection reduces the value of a house. Give one reason why this conclusion may be invalid.

c



5. Daily total rainfall $r$ (mm) and sunshine $s$ (hours) in Leuchars on 11 randomly selected days in August 1987, from the large data set.

| r | 0 | 6.8 | 0.9 | 4.8 | 0 | 21.7 | 1.7 | 4.9 | 0.1 | 2.2 | 0.1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| s | 8.4 | 4.9 | 10.2 | 4.5 | 3.3 | 3.9 | 5.4 | 1.8 | 9.7 | 1 | 4.6 |

The rainfall quartiles are $Q_1 = 0.1,\ Q_2 = 1.7,\ Q_3 = 4.85$. An outlier lies more than 1.5 interquartile ranges above the upper or below the lower quartile.

(a) Show that $r = 21.7$ is an outlier.

(b) Give a reason to (i) include and (ii) exclude this day's readings.

(c) Exclude this day and draw a scatter diagram for the remaining ten days.

(d) Describe the correlation.

(e) Could the relationship between rainfall and sunshine be causal? Explain.

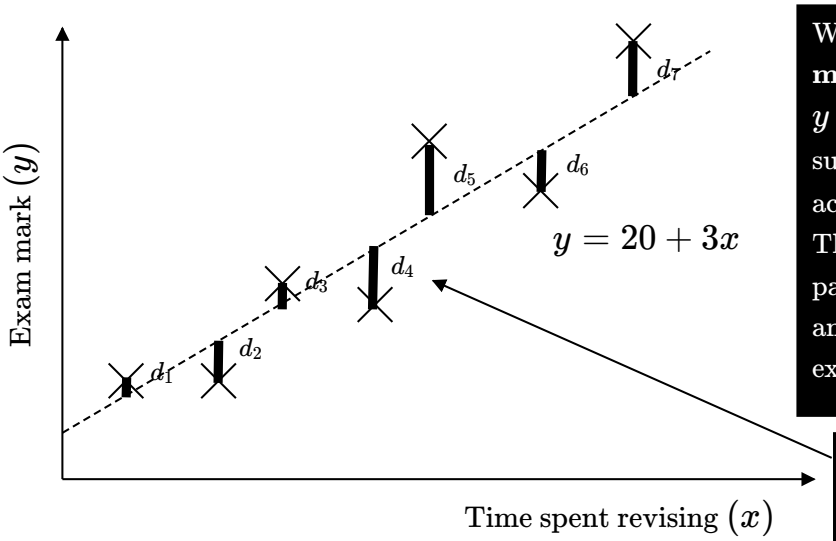
a

b

c

d

e




What we've done here is come up with a **model** to explain the data, in this case, a line $y = a + bx$. We've then tried to set $a$ and $b$ such that the resulting $y$ value matches the actual exam marks as closely as possible. The 'regression' bit is the act of setting the parameters of our model (here the gradient and y-intercept of the line of best fit) to best explain the data.

One type of line of best fit is the **least squares regression line**. This minimises the sum of the square of these 'errors', i.e.

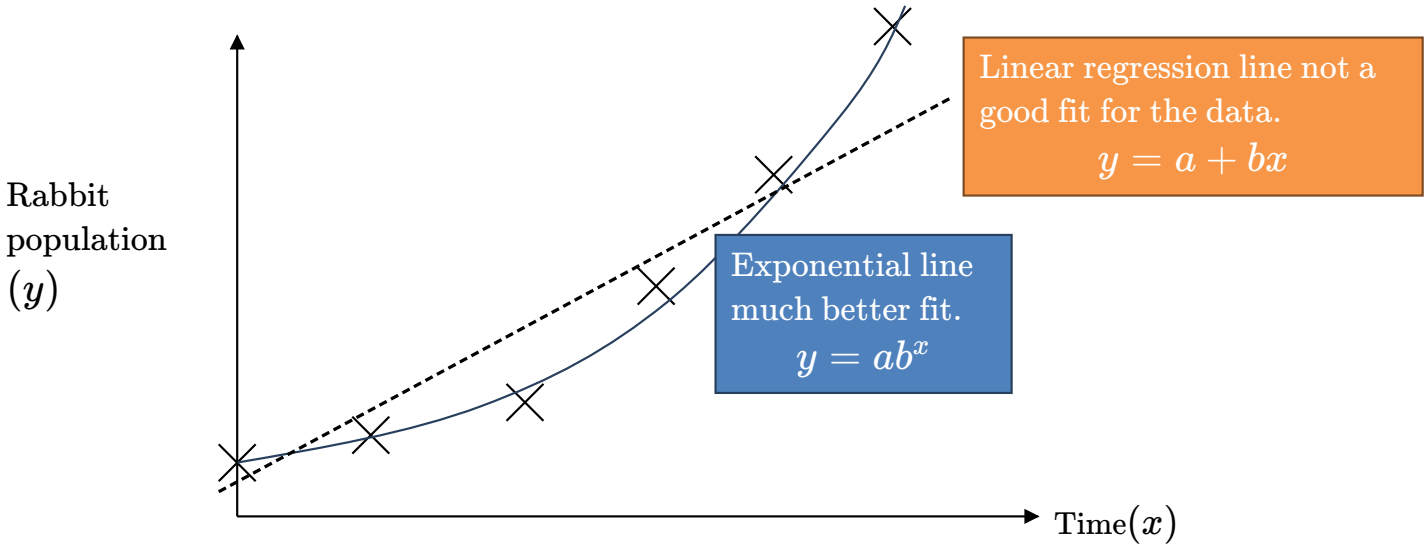
$$d_1^2 + d_2^2 + \ldots = \Sigma d_i^2$$

Part of the reason we square these errors is so that each distance is treated as a positive value.

Unlike in the old S1, you are no longer required to work out the equation of the least squares regression line yourself; you will be given the equation.

I record people's exam marks as well as the time they spent revising. I want to predict how well someone will do based on the time they spent revising. How would I do this?

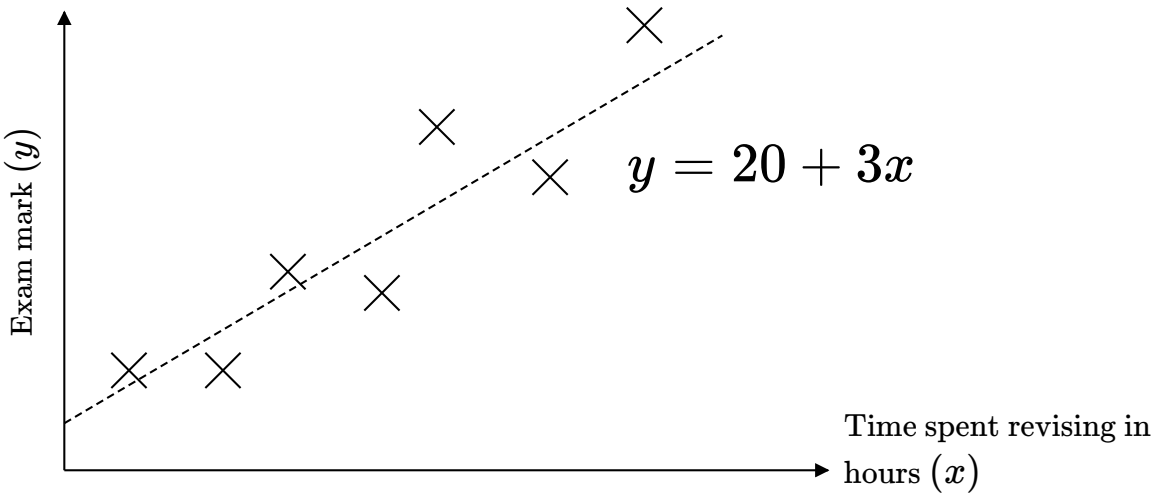





In this chapter we only cover **linear regression**, where our chosen model is a straight line.

But in general we could use any model that might best explain the data. Population tends to grow exponentially rather than linearly, so we might make our model $y = a \times b^x$ and then try to use regression to work out the best $a$ and $b$ to use. **You will do exponential regression in Chapter 14 of Pure Year 1.**

## Interpreting a and b. 11




How do we interpret the gradient of 3?

How do we interpret the $y$-intercept of 20?

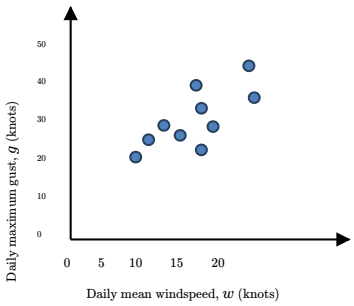
## Example 12

From the large data set, the daily mean windspeed, $w$ knots, and the daily maximum gust, $g$ knots, were recorded for the first 15 days in May in Camborne in 2015.

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **w** | 14 | 13 | 13 | 9 | 18 | 18 | 7 | 15 | 10 | 14 | 11 | 9 | 8 | 10 | 7 |
| **g** | 33 | 37 | 29 | 23 | 43 | 38 | 17 | 30 | 28 | 29 | 29 | 23 | 21 | 28 | 20 |

© Met Office

The data was plotted on a scatter diagram.




(a) Describe the correlation between daily mean windspeed and daily maximum gust.

The equation of the regression line of $g$ on $w$ for these 15 days is $g = 7.23 + 1.82w$

(b) Give an interpretation of the value of the gradient of this regression line.

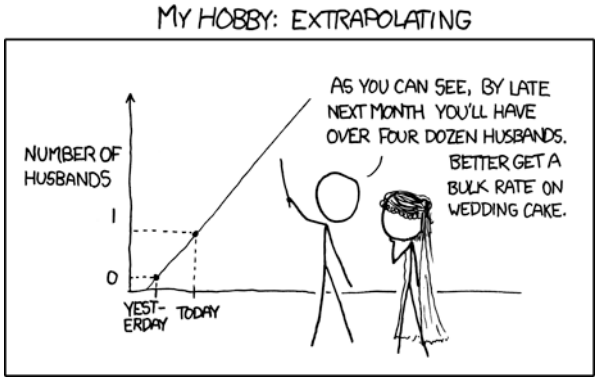
(c) Justify the use of a linear regression line in this instance.

a

b

c

The stronger the (linear) correlation, the more suitable a linear regression line is.

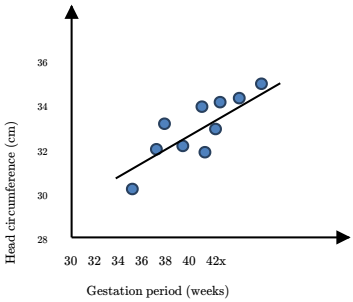


xkcd.com

You should only use the regression line to make predictions for values **of the dependent variable** that are **within the range of the given data**.

Estimating a value inside the data range is known as **interpolating**.
Estimating a value outside the data range is known as **extrapolating**
(as per the cartoon on the left!)

[Textbook] The head circumference, $y$ cm, and gestation period, $x$ weeks, for a random sample of eight newborn babies at a clinic are recorded.
The scatter graph shows the results.
The equation of the regression line of $y$ on $x$ is $y = 8.91 + 0.624x$. The regression equation is used to estimate the head circumference of a baby born at 39 weeks and a baby born at 30 weeks.
(a) Comment on the reliability of these estimates.
A nurse wants to estimate the gestation period for a baby born with a head circumference of 31.6cm.
(b) Explain why the regression equation given above is not suitable for this estimate.




a

b



5. The large data set gives maximum relative humidity $h$ (%) and mean visibility $v$ (decametres) at Heathrow during the first two weeks of September 2015.

| h | 94 | 95 | 92 | 80 | 97 | 94 | 93 | 90 | 87 | 95 | 93 | 92 | 91 | 98 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| v | 2600 | 2900 | 3900 | 4300 | 2800 | 2400 | 2700 | 3500 | 3000 | 2200 | 2200 | 3300 | 2800 | 2200 |

The regression line of $v$ on $h$ is $v = 12700 - 106h$.
(a) Interpret its gradient.
(b) Use your knowledge of the large data set to explain whether there is likely to be a causal relationship.
(c) Why would it be unreliable to predict (i) mean visibility at 100% humidity and (ii) humidity when visibility is 3000 decametres?
(d) State two ways to make better use of the large data set to model the relationship.

a

b

c

d

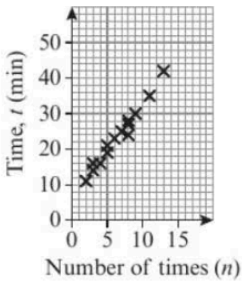
3. Packing times for a mail-order company employee were recorded for a random sample of orders.

| Items (n) | 2 | 3 | 3 | 4 | 5 | 5 | 6 | 7 | 8 | 8 | 8 | 9 | 11 | 13 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Time (min) | 11 | 14 | 16 | 16 | 19 | 21 | 23 | 25 | 24 | 27 | 28 | 30 | 35 | 42 |

(a) Describe the correlation between items packed and time taken. (1 mark)

The regression line is $t = 6.3 + 2.64n$.

(b) Interpret 2.64. (1 mark)




3a

3b

6. James measured spring length $s$ (cm) for attached masses $m$ from 20 to 100 g. The regression line is $s = 44 + 0.2m$.

(a) Interpret 44 and 0.2. (2 marks)

(b) Why would it be unreasonable to use this equation to estimate (i) $s$ when $m = 150$ or (ii) $m$ when $s = 60$? (2 marks)

6a

6b